

Scalability & Future Plan

Date	01 October 2026
Team ID	SWTID-2026-8779
Project Name	PocketSmartAI – AI-Powered Personalized Budget Planning System
Maximum Marks	1 Mark

Scalability & Future Plan:

PocketSmart AI can be scaled to support more users, recommendation requests, product categories, and external platforms. Future improvements will focus on persistent database storage, real-time product data integration, improved AI recommendations, faster processing, enhanced security, and mobile application support.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	AI recommendations depend on external Gemini API availability	Product price and availability may require updated external data	High
2	Current system uses local fallback recommendations	User history may not scale efficiently with more users	High
3	AI analysis depends on available user data	Recommendation generation may be affected when the AI service is unavailable	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for current users	Deploy using scalable cloud infrastructure
2	Data Storage	Basic project-level storage	Integrate a persistent cloud database
3	Performance	Suitable for current workload	Add caching and asynchronous processing
4	Security	JWT-based authentication	Add advanced encryption and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Persistent Database Integration	1–2 Months	Reliable storage of users and recommendation history
Phase 2	Real-Time Product & Service Integration	2–3 Months	More accurate prices and availability
Phase 3	Advanced AI Personalization	3–4 Months	More relevant budget-based recommendations